

Machine learning surveillance of foodborne infectious diseases using wastewater microbiome, crowdsourced, and environmental data

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ABSTRACT

Clostridium perfringens (CP) is a common cause of foodborne infection, leading to significant human health risks and a high economic burden. Thus, effective CP disease surveillance is essential for preventive and therapeutic interventions; however, conventional practices often entail complex, resource-intensive, and costly procedures. This study introduced a data-driven machine learning (ML) modeling framework for CP-related disease surveillance. It leveraged an integrated dataset of municipal wastewater microbiome (e.g., CP abundance), crowdsourced (CP-related web search keywords), and environmental data. Various optimization strategies, including data integration, data normalization, model selection, and hyperparameter tuning, were implemented to improve the ML modeling performance, leading to enhanced predictions of CP cases over time. Explainable artificial intelligence methods identified CP abundance as the most reliable predictor of CP disease cases. Multi-omics subsequently revealed the presence of CP and its genotypes/toxinotypes in wastewater, validating the utility of microbiome-data-enabled ML surveillance for foodborne diseases. This ML-based framework thus exhibits significant potential for complementing and reinforcing existing disease surveillance systems.

1. Introduction

Foodborne illnesses are a pressing concern for public health management, with 2.4 million cases a year in the UK, 4 million in Canada, 48 million in the US, and 600 million worldwide, with half a million human deaths annually (World Health Organization, 2015). These infections result in a substantial economic burden, leading to a \$15–\$60 billion expenditure in the US (Hoffmann and Ahn, 2021). Of particular concern is *Clostridium perfringens* (CP), a Gram-positive, spore-forming aerobic bacterium that inhabits the gastrointestinal tracts of both humans and animals and is responsible for a variety of diseases, including food poisoning, gastrointestinal disorders, necrotizing enterocolitis, gas gangrene, and liver and kidney damage (Dolan et al., 2016). Common clinical manifestations include symptoms such as diarrhea, headache, nausea, vomiting, and stomachache. CP ranks as the second most prevalent foodborne pathogen in Canada (Bélanger et al., 2015). CP contributes to a million cases and accounts for about 1 % of all hospitalizations in the US (Hoffmann and Ahn, 2021). CP causes about 4 million cases across 61 European countries (World Health Organization, 2015).

The surveillance of foodborne diseases to determine their prevalence

and epidemiological characteristics and to monitor the emergence of new or variant pathogens is important for safeguarding public health. This surveillance requires the ongoing collection of case data, which is then analyzed and interpreted so that preventive and/or therapeutic actions such as interventions, vaccinations, and mass drug administration can be taken. The efficacy of this form of disease surveillance relies on the establishment of robust public health networks in which physicians and laboratories report diagnosis records to the relevant public health authorities. Although these practices have been widely adopted globally, they face inherent complexities that are complicated by the intricate hierarchical structures within healthcare systems, coupled with constraints related to time and resources. These challenges often result in the incomplete or unreported documentation of disease cases (Buckee et al., 2018) and the delay between the occurrence of an event and its official publication in regional and national surveillance records (e.g., 1 to 2 weeks) (Ginsberg et al., 2009a; Milinovich et al., 2014).

Google Flu Trends (GFT) is a form of data-driven surveillance that collects queries from online search engines associated with symptomatic signals of human diseases (Milinovich et al., 2014). GFT can be used to monitor the prevalence of influenza-like illnesses within a given population, reducing reporting delays to approximately one day. Due to

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benefits like this, this type of digital surveillance system, which can tap into crowdsourced data from platforms such as Twitter, Yelp, and Yahoo, (Tao et al., 2021a) has garnered significant attention. In particular, recent research has demonstrated the feasibility of computerized epidemiological surveillance using Google web searches and location data to inform hotspots, particularly restaurants with frequent lapses in food safety, which may contribute to the transmission of foodborne illnesses (Sadilek et al., 2018). Environmental conditions are also associated with the emergence and scale of foodborne infectious diseases (Stephen and Barnett, 2016), with CP cases having been linked to climatic factors such as the wind-chill temperature, humidity, and sunshine rate (Oh et al., 2021). In addition, increased infection risks often coincide with precipitation because runoff from agricultural/livestock areas can increase non-point source contamination in the environment, while the temperature affects the growth and replication

of bacterial pathogens, increasing the likelihood of food contamination.

Wastewater-based epidemiology (WBE) samples and analyzes biological and chemical residues in wastewater, enabling inferences to be made about a population's consumption, exposure, and health risks related to these residues (Sims and Kasprzyk-Hordern, 2020). Initially introduced to monitor pathogenic (e.g., poliovirus) and hazardous agents (National Academies of Sciences, Engineering, and Medicine, 2023), WBE has gained prominence due to its ability to predict temporal trends in infectious diseases, as illustrated by the recent COVID-19 pandemic, during which viral loads in domestic wastewater were analyzed (Amman and Bergthaler, 2022; Amman et al., 2022; Karthikeyan et al., 2022). Additionally, the isolation and characterization of foodborne pathogens (e.g., *Salmonella*) in wastewater samples have revealed the presence of various virulence factors, suggesting their potential utility as indicators of the overall disease burden and toxinotypes

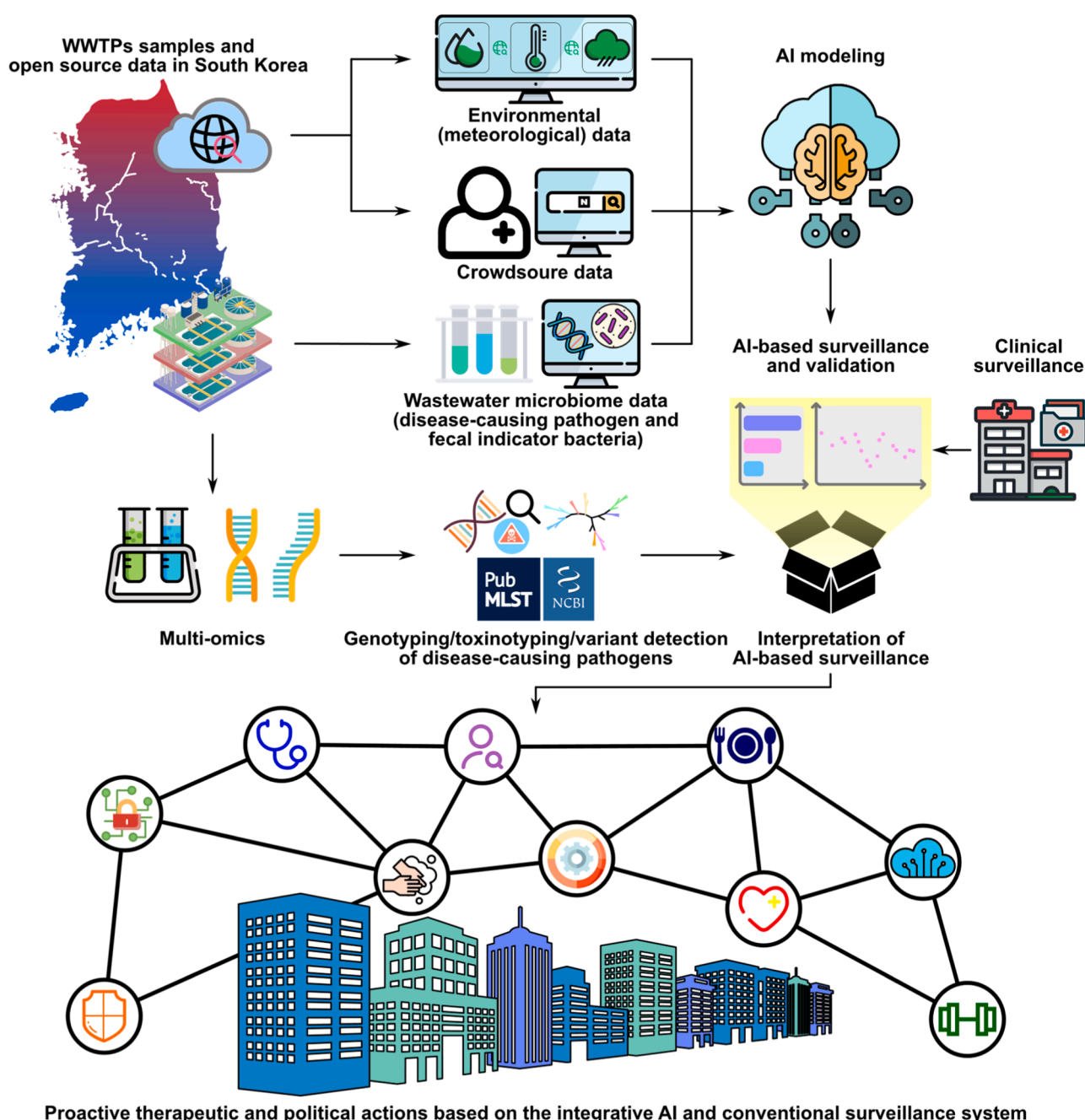


Fig. 1. ML modeling framework established in the present study.

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In the present study, we developed a data-driven machine learning (ML) modeling framework for the surveillance of CP foodborne illnesses (actual CP cases as a target variable). This approach involved amalgamating extensive municipal wastewater, environmental, and crowdsourced web datasets (Fig. 1) as input variables. In doing so, we created a comprehensive and structured dataset that combined high-throughput sequencing (HTS) data for wastewater microbiomes, 40 meteorological indices, and 9 online search queries associated with CP illness symptoms. We hypothesized that this multifaceted dataset containing a variety of input variables may be associated with (i.e., influence and/or respond to) CP cases as the target variable. Given the absence of explicit relationships among the variables reported in literature, our study harnessed ML modeling to avoid a priori assumptions, such as the simple linear relationships that have been assumed in previous studies. This approach offers flexibility in the handling of complex relationships, e.g., non-linear and hierarchical. Additionally, we implemented model selection, hyperparameter tuning, and data integration, preprocessing, and normalization as optimization strategies to enhance the prediction accuracy for CP cases. In addition to the targeted HTS data used to assess the temporal trends in CP cases (i.e., 16S rRNA genetic data), we conducted non-targeted metagenome and metatranscriptome sequencing analysis to infer the genotypes/toxinotypes of CP present in wastewater, thus validating the results of the proposed ML-based disease surveillance framework.

2. Materials and methods

2.1. Study setting and data collection

In this study, we collected environmental (meteorological factors), crowdsourced (keyword search trends related to CP symptoms) and wastewater (microbiome) data in Korea during the investigation period from February 15, 2018 to July 4, 2019. Daily meteorological data were retrieved from the automated synoptic observation system run by the Korean Meteorological Administration. The meteorological data (Table S1) was obtained from 105 observatory stations across the country via <https://data.kma.go.kr> (publicly available) during the investigation period. Crowdsourced data, consisting of relative counts for web search keywords related to CP symptoms (as detailed in Table S2), were acquired from NAVER DataLab (<https://datalab.naver.com>). NAVER DataLab provides relative query counts (ranging from zero to one hundred) with no specific user information (i.e., no personal identity or IP address information) in user-defined time periods, similar to Google Trends (Adams et al., 2023). Daily relative count data across the country were retrieved during the investigation period. Actual CP cases were obtained on a weekly basis from the official platform of the (Korea Disease Control and Prevention Agency KDCA, 2024) (<https://npt.kdca.go.kr>) during the investigation period. These data, reported by KDCA, were based on clinical laboratory diagnosis criteria established by the agency. A total of 66 mixed liquor samples (100 ml each) were collected from three full-scale WWTPs located in Gyeonggi Province (Fig. S1) during the investigation period. These WWTPs had daily treatment capacities ranging from 4000–174,000 m³/d. Sampling occurred at various points in the treatment train, as shown in Fig. S1. Single samples were collected from the three WWTPs over the investigation period (February 20, April 17, May 16, July 5, August 14, September 21, October 17, and December 17 in 2018 and February 6, March 22, April 12, and July 4 in 2019), yielding 36 samples. Triplicate mixed liquor samples were taken on February 20, 2018 to compare differential CP counts across redox conditions and between the influents and bioreactors, yielding additional 30 samples. Metagenome and metatranscriptome samples were further collected on September 14–17, 2020 from the influents (100 ml mixed liquor sampled) of the three WWTPs. The sampling and data generation/acquisition schemes for all input and target variables used in the predictive ML modeling in this

study are detailed in the Supplementary material: environmental (Table S1), crowdsourced (Table S2), wastewater (Figs. S1 and S2), and CP case (Fig. S2) data.

2.2. High-throughput sequencing and bioinformatic analysis of wastewater microbiomes

DNA extraction from wastewater was conducted using a DNeasy PowerSoil kit (Qiagen, Carlsbad, CA, USA). The quality and quantity of the DNA were checked using a NanoDrop spectrophotometer. PCR amplification was employed to generate 16S rRNA gene amplicons using the hypervariable region (V3 to V5), and these were sequenced using a MiSeq sequencer (Oh and Kim, 2022; Park and Seungdae, 2020). Wastewater samples from the three WWTPs in September 2020 were filtered using a 0.22-μm Sterivex filter (Millipore) for metagenome and metatranscriptome sequencing. DNA was extracted following the same protocol used for 16S rRNA gene sequencing. DNA libraries were constructed using a TruSeq DNA Nano Sample Preparation Kit (Illumina, CA, USA). These libraries were purified and quantified using the qPCR Quantification Protocol Guide (San Diego, CA, USA). Metatranscriptome sequencing was conducted as described previously (Oh et al., 2014). In brief, total RNA was extracted using the Trizol reagent protocol from residues collected on the filters. The extracted RNA was subjected to purification steps using the RNase-Free DNase Set and Proteinase K (Qiagen, Carlsbad, CA, USA). RNA libraries were constructed using a TruSeq mRNA Sample Preparation kit (Illumina, CA, USA). mRNA was enriched via polyA selection, and fragmented RNA was converted into first-strand cDNA with SuperScript II reverse transcriptase (Invitrogen) using random hexamers. The second-strand cDNA was synthesized with DNA Polymerase I and RNase H. End-paired cDNA was purified and enriched to construct the final cDNA libraries, which were quantified using the qPCR Quantification Protocol Guide (San Diego, CA, USA) and qualified using TapeStation D1000 ScreenTape (Agilent Technologies, Waldbronn, Germany). Metagenome and metatranscriptome sequences were generated from DNA and cDNA aliquots using an Illumina NovaSeq sequencer.

The 16S rRNA gene sequences were analyzed as described previously (Nguyen and Oh, 2023) using the MiSeq SOP pipeline (Kozich et al., 2013). Initial quality control included filtering sequences with a Q = 20 Phred quality score, while chimera sequences were removed using chimera.vsearch. Sequences were clustered as OTUs using OptiClust with default parameters and then taxonomically classified using classify.seqs. The analysis produced an OTU table with normalized counts across datasets. Metagenomic and metatranscriptomic datasets were trimmed using a Q = 20 Phred quality score cutoff with Trimmomatic 0.39 (Bolger et al., 2014). The trimmed reads were searched against the multi-locus sequence typing (MLST) database for CP (PubMLST (Jolley et al., 2018)) using BLASTn with a sequence identity of > 97 % and a query length coverage of > 85 %. Both single and merged reads were subject to phylogenetic analysis.

2.3. Development of the ML modeling framework

The input (microbiome) and target (CP case) variables were transformed on a daily basis, consistent with the other variables (environmental and crowdsourced data). The wastewater microbiome data were converted into daily values through spline interpolation using the `splinefun` function in R. A three-point moving average was then applied using `rollapply` in R, as the methods are commonly performed in temporal trend analysis (Ai et al., 2021; Rauch et al., 2022; Zheng et al., 2022). To obtain daily target variable data, the original weekly values were divided by seven. All daily input and target variables were subsequently integrated into a combined dataset, as illustrated in Fig. S2. The ML modeling was based on a microbiome data-driven ML modeling pipeline previously developed (Kim and Oh, 2021; Kim et al., 2021; Oh and Kim, 2021; Oh et al., 2022; Wijaya et al., 2023) using SciKit-Learn

0.23.2 (Pedregosa et al., 2011) following modifications for the purposes of the present study. A total of eight ML regression models were tested for their ability to predict the target variables: ordinary linear regression (OLR), linear regression with lasso regularization (LRLR), linear regression with ridge regularization (LRRR), support vector regression with linear kernel (SVRL), support vector regression with radial basis function (SVRRBF), random forest regression (RFR), AdaBoost regression (ABR), and gradient boost regression (GBR). The selection of these models was based on their widespread use and their varying levels of complexity in terms of algorithm design and the interpretability of their predictive regression. The input data were preprocessed using min-max scaling in Scikit-Learn. Hyperparameter optimization was conducted via five-fold cross-validation with the training data using StratifiedKfold and RandomizedSearchCV. The performance of each model with its optimized hyperparameters was evaluated using the test dataset, which contained data that were not used during model training, based on two performance metrics (MSE and R^2). To interpret the ML-based predictions, two distinct feature importance analyses were conducted using GBR and LRLR, respectively. The OVR method (Wijaya et al., 2023) was used to determine feature importance for LRLR and the SHAP technique (Lundberg et al., 2020) was employed for GBR using SHAP20 V. 0.34.0 (<https://github.com/slundberg/shap>).

3. Results

As illustrated in Fig. 1, the ML modeling framework was initiated with a monitoring campaign that took wastewater samples from three full-scale wastewater treatment plants (WWTPs) in Korea. Details of the treatment processes, design parameters, and sampling are described in Fig. S1a. A total of 66 mixed liquor samples were taken from February 2018 to July 2019 and subjected to 16S rRNA gene sequencing. Taxonomic analysis revealed the presence of OTU2058 in the wastewater microbiome, which was related to *Clostridium*. A sequence homology

search using BLASTn revealed that OTU2058 was most closely related to *C. perfringens* with $a > 99.5$ % identity (Fig. 2 and Table S3), compared to < 97 % for other *Clostridium* spp. It is worth noting that its closest relatives have previously been observed in clinical specimens obtained from individuals diagnosed with conditions such as gas gangrene, antibiotic-associated diarrhea, and gastroenteritis.

The continuous presence of CP and the FIBs was also observed in the microbiome monitoring campaign and bioinformatics analysis over the study period. In particular, the CP population was consistently observed with over > 95 % occurrence in the wastewater microbiomes over time. Although the occurrence patterns for OTU2058 were similar across the three WWTPs and their redox tanks (Fig. S1b), it was 2–4 times more abundant in the influents than in the biological tanks. Taxonomic analysis also identified the 100 % occurrence of two populations of *Escherichia* (OTU0268) and *Bacteroides* (OTU0206) over time. These genera are commonly found in waste-associated samples and often used as fecal indicator bacteria (FIB) (Maal-Bared et al., 2023). Phylogenetic analysis revealed a close phylogenetic relationship between the two FIB OTUs and those found in human samples such as feces and blood (Figs. S2 and S3).

In addition to the wastewater microbiomes, our dataset was enriched with input data sourced from urban environments and digital social networks. A comprehensive set of 40 environmental variables (Table S1) was acquired from the Korean Meteorological Administration. This dataset consisted of ten meteorological factors, each described using four statistical values (mean, minimum, maximum, and 1 standard deviation). Additionally, we collected historical records for web search keywords associated with CP symptoms (Table S2). CP cases were obtained from the Korea Disease Control and Prevention Agency and used as a target variable. Temporal data for the variables were preprocessed and collectively structured into the integrative dataset (Fig. S2). The integrative data were used for the ML modeling framework, which was designed to predict the actual number of cases of CP using supervised

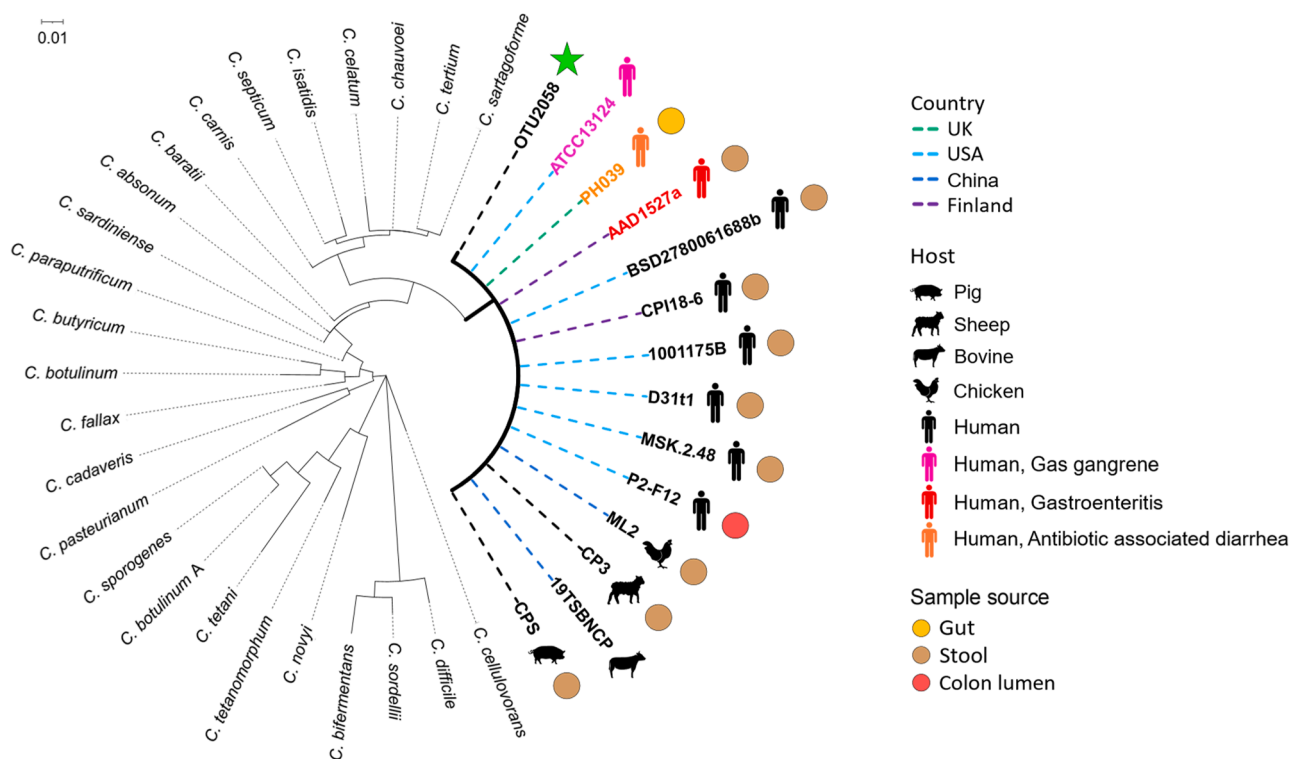


Fig. 2. Phylogenetic relationship of the *Clostridium* populations. The phylogenetic tree was generated based on 16S rRNA genes using the maximum likelihood method with the Tamura-Nei model. CP references are denoted by the bolded branch lines and show $a > 99.5$ % identity with OTU2058 (marked with a green star). The accession numbers for the CP references are presented in Table S3. The country, host, and specimen information for the CP references is indicated using dotted lines, symbols, and circles, respectively.

learning algorithms in a total of eight models. These models included four linear and four non-linear/hierarchical models and encompassed a range of model complexity and interpretability. We conducted an exhaustive exploration of a range of hyperparameters for each model, optimizing them via a random search option with five-fold cross-validation. The parameter tuning process was conducted with the training dataset (70 % of the original data) and the prediction accuracy of the models was subsequently evaluated using the test set (the remaining 30 % of the dataset). Prediction accuracy was assessed using both the coefficient of determination (R^2) and the mean-square error (MSE), respectively (Fig. 3a, b). The non-linear models ($R^2 = 0.32$ – 0.80) outperformed the linear models ($R^2 = 0.10$ – 0.63) in terms of prediction accuracy, while the extent of data integration was also found to significantly affect the results (Fig. 3a, b). In particular, the highest prediction accuracy ($R^2 = 0.20$ – 0.80) was achieved when integrating all three types of data (ECW), followed by EC (0.10 – 0.53) and E (0.21 – 0.36). The results using the MSE followed the same trends.

Of the eight models, GBR and LRLR exhibited the highest accuracy. To account for environmental (i.e., in-sewer) factors potentially affecting microorganism levels in the wastewater (Maal-Bared et al.,

2023; Mitranescu et al., 2022), normalization based on the FIB (*Escherichia* and *Bacteroides*) observed in Fig. 1 was conducted. In particular, the CP abundance was normalized by both the operational taxonomy units (OTUs) of the two FIB and the genera that the OTUs belonged to. The approach resulted in four normalization methods for CP abundance by dividing it by the abundance of OTU0268 (ECWN1), all *Escherichia* (ECWN2), OTU0206 (ECWN3), and all *Bacteroides* (ECWN4), respectively. The effect of this normalization process on the prediction performance was significant, leading to greater accuracy for both GBR and LRLR with the normalized results ($R^2 = 0.75$ – 0.78 and 0.42 – 0.64 , respectively) (Figs. 3c and S5) compared to those with the non-normalized results ($R^2 = 0.34$ – 0.72 and 0.28 – 0.38 , respectively) (Fig. 3a).

Hyperparameter tuning was performed to optimize each model. For example, GBR with the best performance was optimized by adjusting the number of decision trees and the learning rate. Increasing the tree number increases the expressiveness and power of the model, but an excessive increase can lead to overfitting issues. The learning rate regulates the relative contribution of each estimator to the overall prediction, and a rate that is too high or too low can cause overshooting,

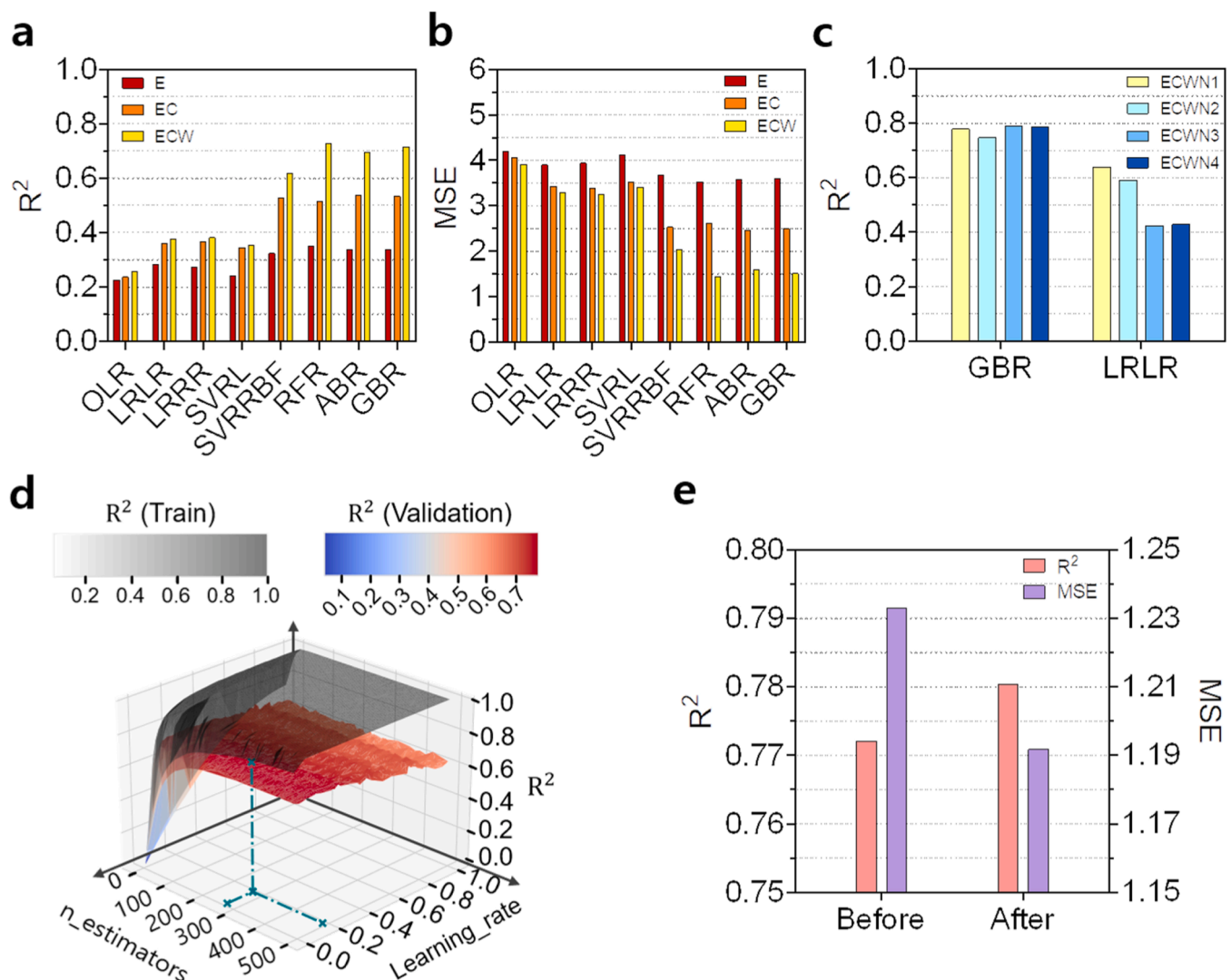


Fig. 3. ML modeling for the prediction of actual CP cases. Prediction accuracy of ML models in terms of (a) R^2 and (b) MSE. Eight models were used: OLR, LRLR, LRRR, SVRL, SVRRBF, RFR, ABR, and GBR. The modeling was conducted with three datasets: environmental (E), environmental and crowdsourced (EC), and environmental, crowdsourced, and wastewater microbiome (ECW). (c) Modeling performance with four normalization approaches (ECWN1 to ECWN4) using the ECW dataset. (d) Modeling performance of GBR with a range of hyperparameter values. (e) Comparative modeling performance of GBR before (default) and after hyperparameter tuning. Note the high prediction performance after hyperparameter tuning (i.e., an increase in R^2 and a decrease in MSE).

instability, or poor performance (Boehmke and Greenwell, 2019). Although a very high performance was observed at increasing hyperparameters with the training data, which indicates potential overfitting for the range of parameters presented in Fig. 3d, the tuning process identified 341 trees and a learning rate of 0.16 as the optimal hyperparameters with the test data, which fell within the typical optimum range (Boehmke and Greenwell, 2019). The improvement in the prediction accuracy due to hyperparameter optimization is evident from the R^2 and MSE values presented in Fig. 3e. Collectively, these results highlight the importance of optimization strategies such as data integration, normalization, model selection, and hyperparameter tuning. These efforts led to the development of a robust predictive model ($R^2 = 0.78$) that was able to capture the dynamic temporal trends of actual CP cases, including during normal (weeks 1–55) and peak (weeks 56–73) periods (Fig. 4).

The results of our ML modeling indicated that the 50 input variables from the three independent data sources were useful predictors for assessing the number of CP cases. Artificial intelligence (AI) models are often perceived as black boxes, lacking explicit explanations for the predictions they produce, thus limiting their practicality in various fields (Ghannam and Techtman, 2021; Long et al., 2022; Wijaya et al., 2023). To address this, we conducted a feature importance and sensitivity analysis, a commonly used technique in explainable AI (XAI), to quantify the relative contribution of each variable to the ML-based predictions. The global sensitivity analysis results for the GBR model using the Shapely additive explanations (SHAP) method are presented in Fig. 5a. The SHAP approach, developed based on cooperative game theory, assesses the relative contribution of each feature within a cooperative coalition to the overall prediction (Lundberg et al., 2020). Among the 50 input variables, the CP feature had a SHAP value of 1.4, which was significantly higher than any other (< 0.2). In addition to the wastewater-derived CP feature, the top ten features with high SHAP values included six environmental (0.05 to 0.13) and three crowd-sourced (0.08 to 0.20) features.

A local explanation of the ML predictions is presented in Fig. 5b, with the aim to assess whether each feature positively or negatively influenced the predictions. The red dots on the right axis, with positive SHAP values, indicate a positive association with high number of CP cases. The local sensitivity analysis emphasized that the wastewater-derived CP count was a strong predictor of high CP cases. In parallel with the SHAP approach, feature weights were evaluated using LRLR, employing the

one-vs.-rest (OVR) strategy. The analysis confirmed the significant contribution of CP data with LRLR (a feature weight of 5.6), which was consistent with the GBR results (Fig. 5a). The consistency between the independent SHAP and OVR analyses provided valuable insights into how input features affect the prediction results, facilitating their interpretation.

Time-series sampling and sequencing of the wastewater revealed the continuous presence of CP populations that are phylogenetically associated with human hosts. We subsequently sought to characterize the genotype/toxinotype diversity of CP in wastewater using metagenomic and metatranscriptomic sequencing. Metagenomes and metatranscriptomes were recovered from the wastewater, resulting in 19–40 Gb of paired reads per dataset. Following the quality control of the reads, a sequence homology search was conducted using BLASTn against the multi-locus sequence typing (MLST) database for CP. The search results uncovered genomic and transcriptomic signatures of CP in the microbiomes of WWTP2 and WWTP3, revealing the presence of genetic (*colA* and *pfoA*) and transcript (*cpa* and *colA*) sequences encoding virulence factors/toxins associated with CP (undetectable in those of WWTP1). Two gene (WWTP2-MG and WWTP3-MG) and two gene transcript (WWTP2-MT and WWTP3-MT-M) sequences were identified from WWTP2 and WWTP3, respectively.

To gain further insight, a phylogenetic tree was constructed with four read sequences with closely related references obtained from the NCBI database (Fig. 6). Many of the reference strains associated with human biospecimens (i.e., from the gut, stool, and colon) that were clinically diagnosed in association with conditions such as gas gangrene, gastroenteritis, and diarrhea exhibited a high degree of similarity ($>97.6\%$) to the reads recovered from the wastewater microbiomes. In particular, three reference strains (PH039, BSD2780061688b, and D31t1) consistently clustered within the same lineage as the CP populations retrieved from the wastewater microbiomes. Although the genomic assembly of CP was not feasible due to limited metagenome/metatranscriptome coverage, our omics-enabled investigation allowed for the identification of CP populations associated with known genotypes/toxinotypes in wastewater consistently observed at the DNA and RNA levels. This supports earlier findings of human-associated CP observed through 16S rRNA gene sequencing over time (Figs. 2 and 3).

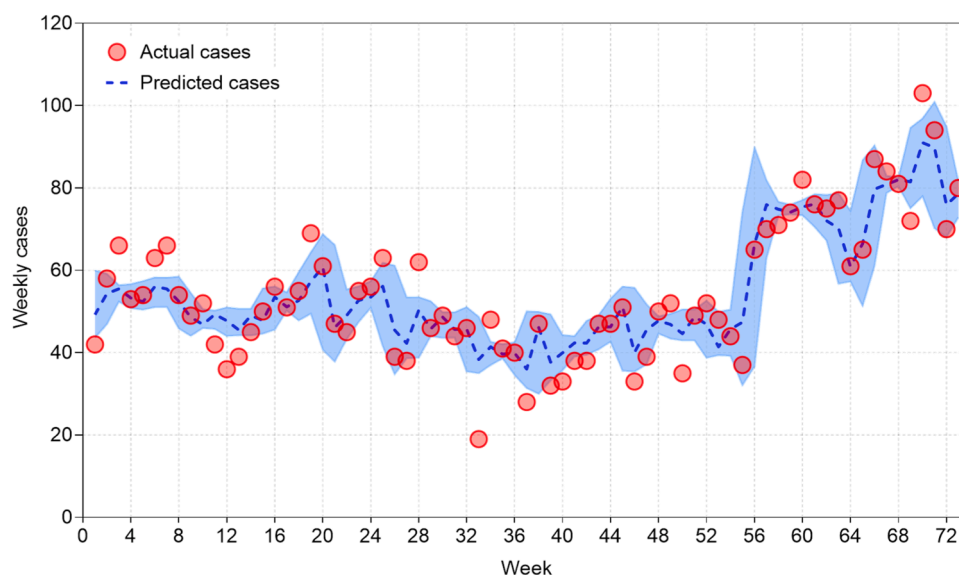


Fig. 4. ML modeling prediction for actual CP cases over time. Actual (red circle) vs. predicted (blue dotted line) cases over time. The blue band represents the 95 % confidence interval for the predictions.

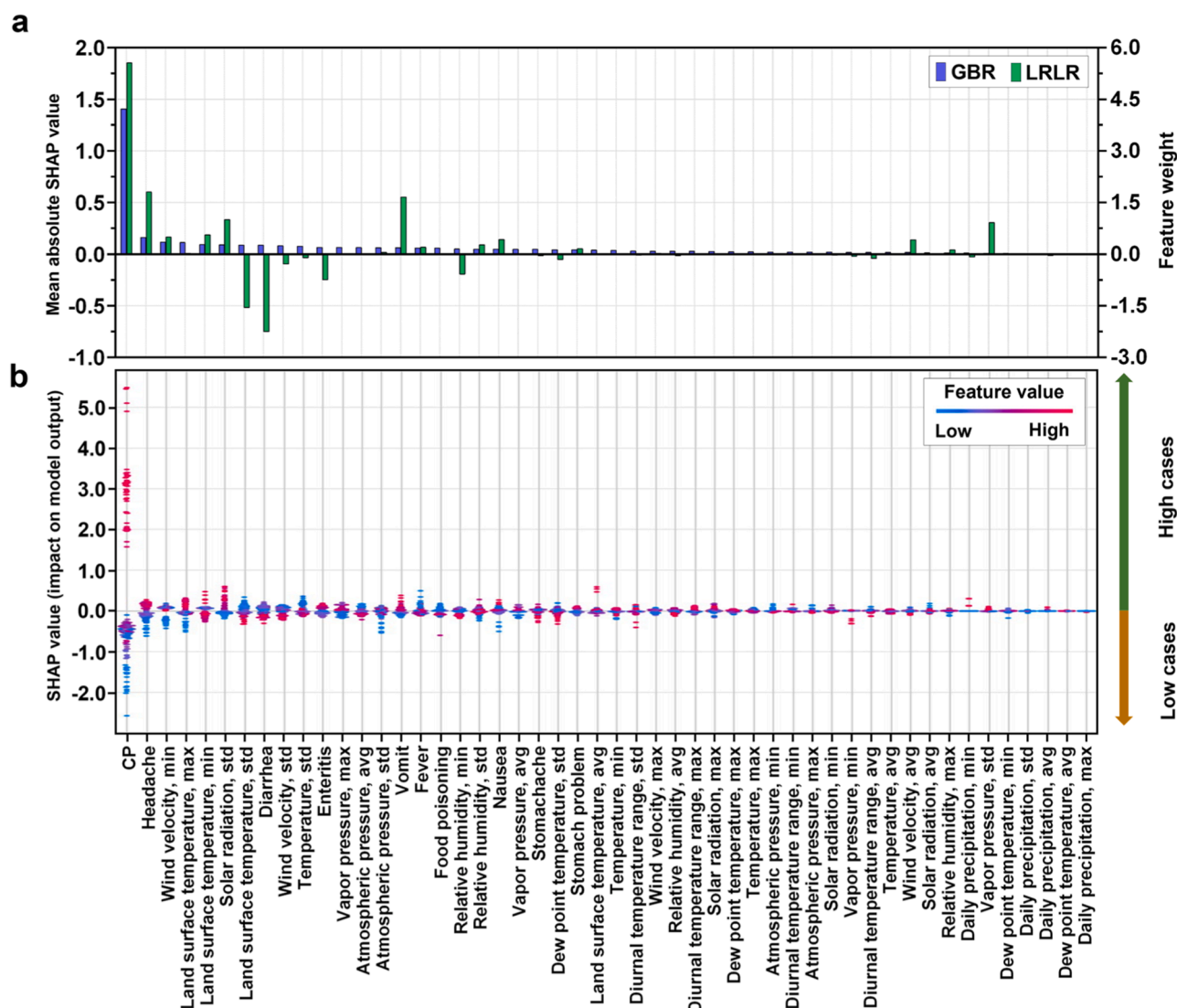


Fig. 5. Input variables ranked based on feature importance values. (a) Mean absolute SHAP and feature weight values estimated using GBR and LRLR, respectively. (b) Dot plots for the features. Red dots plotted with positive SHAP values indicate the relative importance of a feature for ML prediction associated with high CP cases. The red dots for CP (OTU2058) with high SHAP values are strongly associated with high actual CP cases. The ML modeling was carried out with ECWN1 normalization. Min, max, avg, and std indicate the minimum, maximum, mean, and 1 standard deviation, respectively, of the environmental data.

4. Discussion

The present study successfully established an AI-based framework for disease surveillance that distinguishes itself from previous modeling efforts through the incorporation of more extensive and diverse input predictors, particularly the inclusion of wastewater microbiome data. Additionally, it utilized highly optimized and customized AI models to improve upon the performance of previous methodologies. To predict incidence cases/rates of foodborne illness, previous research has predominantly relied on single-sourced input data, either from crowd-sourced sources using search queries and locational/historical logs (Sadilek et al., 2018; Tao et al., 2021a), environmental sources utilizing meteorological data (Bian et al., 2022; Miller-Pierce and Rhoads, 2019; Mitranescu et al., 2022; Stephen and Barnett, 2016), or wastewater sources focusing on pathogen occurrence (e.g., *Salmonella*) (Kuhn et al., 2023). These studies exhibit prediction accuracies of $R^2 = 0.27$ – 0.70 . In contrast, the present study achieved a higher prediction power ($R^2 = 0.78$), largely due to the incorporation of CP microbiome input features, which provided a more direct reflection of population health. The XAI results revealed that the CP feature made the largest contribution to the

AI-based predictions (Fig. 5). Furthermore, the study demonstrated that high prediction performance was associated with the dataset that integrated microbiome data (ECW), which outperformed datasets that lacked microbiome information (i.e., < 0.4 for EC or E) (Fig. 3a). Collectively, these findings provide robust evidence supporting the pivotal role of microbiome data as crucial input variables for disease surveillance.

It is worth noting that prior research predominantly relied on single/simple models such as logistic or Poisson regression (Bian et al., 2022; Sadilek et al., 2018; Stephen and Barnett, 2016) without systematic evidence that these models were the optimal choice. In contrast, the AI modeling framework in this study is highly flexible and offers a wide range of model choices, including complex hierarchical and non-linear models in addition to the typical linear options. From this diverse set of models, the best-performing model can be chosen (Fig. 3a, b) for specific purposes. Furthermore, the selected model can be fine-tuned using the optimization strategies established in the framework (Fig. 3d, e). This flexibility in model selection addresses a common challenge faced by many researchers who have numerous modeling options but do not necessarily know which model best suits their

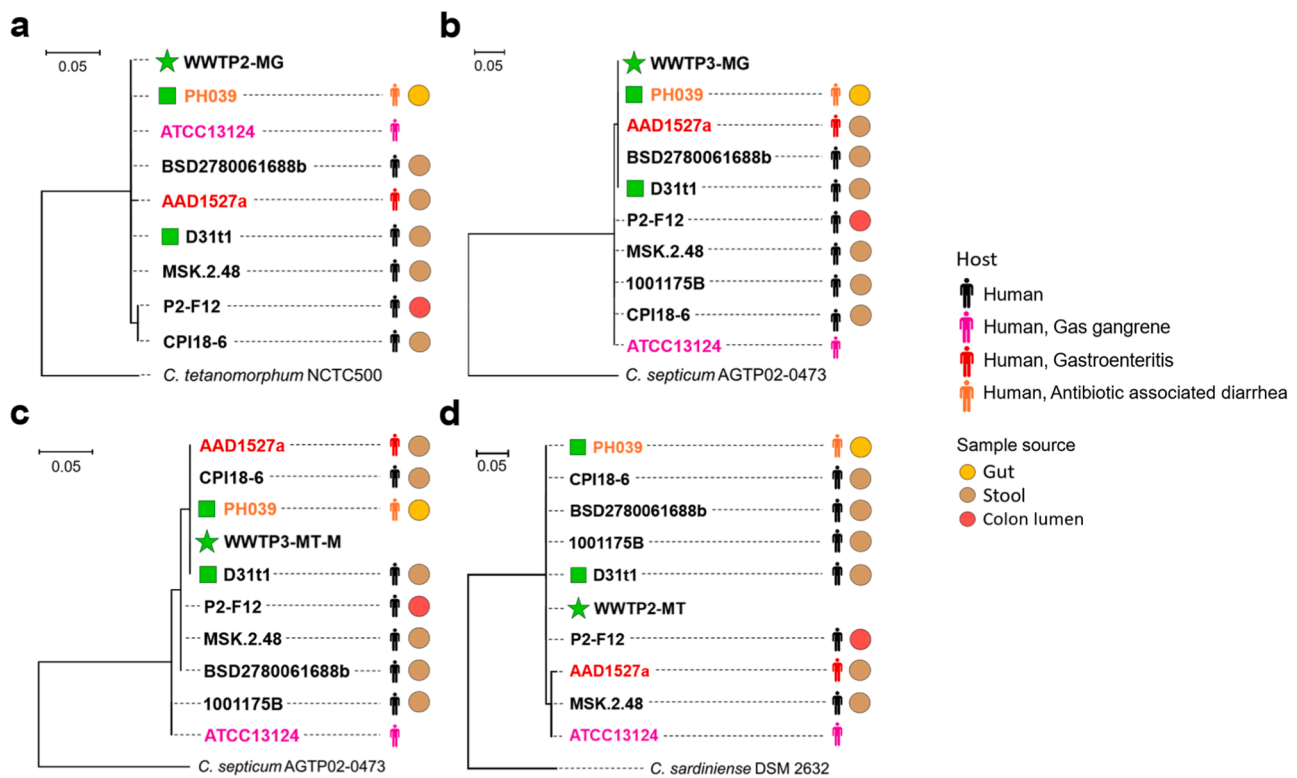


Fig. 6. Phylogenetic tree of the toxin gene/transcript sequences recovered from the metagenomic and metatranscriptomic datasets: (a) *pfoA*, (b and c) *colA*, and (d) *cpa*. Sequences recovered from influent wastewater microbiomes were labeled according to their sampling location (WWTP2 and WWTP3) and source (metagenome [MG] and metatranscriptome [MT]). Paired reads were merged into a longer read (WWTP3-MT-M). The metagenomic/metatranscriptomic sequences shared $a > 97$ % nucleotide identity with CP references linked to human hosts, whereas they showed $a < 85$ % identity with other *Clostridium* species (outgroups). Accession numbers are presented in Table S3.

objectives. In the present study, GBR exhibited the highest performance. GBR often outperforms other commonly used decision-tree models (e.g., RF) because it builds trees sequentially to correct errors from preceding trees, reducing the risk of overfitting, while RF only employs independent trees (Ebrahimi et al., 2019; Morvan et al., 2022). This work also highlighted the importance of hyperparameter tuning and input data normalization in enhancing the AI prediction of the number of disease cases, as has been reported previously (Ghannam and Techtman, 2021; Maal-Bared et al., 2023). Overall, the proposed AI modeling framework derived from integrative data represents a methodological advance in dataset building, preprocessing, normalization, model selection, and hyperparameter tuning.

In contrast to mechanistic models, data-driven models often face challenges associated with the poor explainability of their prediction results and are thus often perceived as black boxes. However, the AI modeling framework developed in the present study addressed this limitation by enabling the quantitative ranking of features based on their relative contribution to the predictions. These highly ranked features provided valuable insights into their influence on the outcomes, enhancing the interpretability of ML prediction results. The utilization of ML modeling with global and local sensitivity analyses uncovers dynamic and complex associations among a number of variables in the biosphere, while the relationships may not have been elucidated using conventional hypothesis testing and mechanistic or mathematical models (Kim and Oh, 2021; Kim et al., 2021; Long et al., 2022; Oh and Kim, 2021; Oh et al., 2022; Wijaya et al., 2023). We recommend the construction of more comprehensive datasets that include environmental microbiomes from various sources (e.g., wastewater, soil, air, and natural water) to provide a more accurate reflection of the health status of urban populations and improve disease surveillance efforts. In this respect, the proposed ML modeling framework with XAI holds great

potential for investigating complex epidemiological challenges and gaining insights into the environmental factors associated with disease epidemiology.

CP occurs more prevalently in warm weather conditions such as spring and summer because these conditions are conducive to its growth in inadequately stored or handled food. The onset of syndromic symptoms such as diarrhea and abdominal pain typically occurs within a day of infection or consumption of the contaminated food. CP has a rapid doubling time, often <10 min (Gohari et al., 2021), allowing for the swift development of symptoms, which can persist for up to a week. While normal fecal samples contain $<10^4$ – 10^5 colony-forming units (CFUs) per gram, $>10^5$ – 10^6 CFUs per gram (Dolan et al., 2016) are observed in cases of food poisoning, which can be used for the clinical diagnosis of foodborne illnesses in Korea. A significant quantity of the CP load released into municipal wastewater systems in Korea can reach WWTPs within hours or, at most, within a day, via urbanized sewer networks (Li et al., 2023). Given these epidemiological circumstances, crowdsourced data (e.g., keyword searches for CP symptoms conducted by infected humans) and wastewater-derived CP data (e.g., infectious agents released from patients) could be considered responsive factors to CP-induced illnesses. While the keywords associated with CP symptoms served as useful input variables in the present study, we recommend that future studies focus on precisely selecting web search queries that are more specific and responsive to particular diseases, as attempted in Ginsberg et al. (2009b), Sadilek et al. (2018), Tao et al. (2021b). On the other hand, the environmental data used in this study (e.g., temperature and sunlight) could be regarded as influential factors.

One of the primary objectives in disease surveillance is the identification of disease-causing agents that circulate in a given area. The analysis of pooled wastewater samples is an effective way to identify these agents over an entire sewershed area. Traditional methods for

identifying disease agents include cultivation using selective agar media (mCP), toxinotyping (RPLA), and polymerase chain reaction (PCR) for the detection of the presence of marker genes such as *cpa*. While these culture-dependent techniques have been officially adopted in many countries, they have limitations in terms of characterizing specific genotypes/toxinotypes. The present study revealed the presence of CP using the high-throughput sequencing of the 16S rRNA gene as a targeted genetic marker in wastewater samples. This finding was then validated by the detection of CP toxin gene/transcript sequences (*cpa*, *pfoA*, and *colA*) using non-targeted HTS technologies at both the DNA and RNA levels. The genetic determinants observed in this study are known to be conserved on the chromosome of CP (Kiu and Hall, 2018; Gohari et al., 2021). CP produces CPA (α -toxin), a zinc-containing phospholipase C enzyme, that is potentially associated with conditions such as gangrene, enterotoxemia, and gastrointestinal disease (Camargo et al., 2022). This enzyme hydrolyzes phosphatidylcholine and sphingomyelin in the structure of the plasma membrane, facilitating its interaction with the receptors (Gohari et al., 2021). Perfringolysin O (θ -toxin) functions as a pore-forming toxin, contributing to the destruction of host tissue and preventing cell lysis by the host's immune system. PFO exhibits cytotoxic effects on macrophages and polymorphonuclear leukocytes, playing a role in lysing endosome membranes within macrophages and assisting CP in escaping from phagosomes (Gohari et al., 2021). The *colA* gene encodes alpha-clostripain collagenase (κ -toxin), which is associated with intestinal infection because collagen is a major component of the intestinal tissue and membranes of human hosts. The enzymatic hydrolysis of collagen disrupts its structural integrity, potentially leading to tissue necrosis (Kiu and Hall, 2018).

Wastewater-derived information regarding the prevailing types of CP can be of practical significance for healthcare treatment decisions made by clinicians. For instance, antibiotic therapy is a common approach for treating foodborne illnesses and the choice of antibiotic depends on factors such as the severity and type of infection and the antibiotic susceptibility of the agent. However, antibiotic resistance profiles can vary among CP toxinotypes over time and in different geographic areas (Feng et al., 2020). Therefore, we recommend that future studies regularly examine the prevailing CP genotypes in wastewater and characterize the antibiotic susceptibility of wastewater-originating CP populations on a wide spatiotemporal scale in a manner similar to the routine practices conventionally conducted in clinical settings. Because the CP population in wastewater microbiomes can represent the pooled representatives of the given human population with fewer sampling biases, scientific findings obtained from wastewater microbiomes would help local healthcare professionals to develop effective and proactive interventions and therapies against infectious diseases. This approach has proven successful in the context of COVID-19, where sequencing wastewater samples has enabled the early detection of emerging viral variants such as Omicron before clinical testing (Safford et al., 2022). These wastewater-based findings have the potential to contribute to proactive collaboration among stakeholders in affected areas and help healthcare decision-makers and practitioners adjust their strategies in response to infectious diseases in a timely manner.

The use of AI modeling for disease surveillance is a promising strategy, but it has certain challenges and limitations. First, though it is capable of producing large volumes of data, current HTS technology can be time-consuming (e.g., days of run time with MiSeq) and costly. Alternative sequencing technologies such as nanopore sequencing enable real-time genomic surveillance of infectious diseases, offering more time-efficient monitoring (Wang et al., 2021; Wijaya et al., 2024). In addition, from a practical standpoint, we may suggest future research exploring the use of fast and easy-to-use molecular tools, such as multiplex quantitative PCR assays (e.g., hours of run time), that can differentiate between sub-species in CP populations. Although HTS technology with universal bacterial primers enables monitoring of

hundreds of potential human pathogens in wastewater, PCR assays may overcome the limitations of HTS sequencing in the analysis of wastewater samples for surveillance purposes. Their workflow is comparable in terms of time and effort to conventional PCR testing practices that are routinely performed on food samples for the official surveillance of foodborne illnesses in Korea and other countries.

Second, the experimental stage of the AI modeling framework involves the collection and handling of wastewater samples, which needs further standardization and optimization. When interpreting AI modeling results, factors such as the sampling scheme, the population size within the sewershed area under investigation, and in-sewer conditions need to be taken into account. These factors may influence the observed levels of CP. Our results showed a higher abundance of CP in the influents than in the biological tanks, highlighting the importance of the sampling scheme (Fig. S1b). Differences in population sizes between WWTPs should also be considered. Because this study focused on measuring temporal trends in CP abundance rather than spatial comparisons, differences in population size were not a significant concern. Furthermore, in-sewer conditions (e.g., water temperature and infiltration from other sources) can affect the fate of and detectable abundance of CP. To address this, our study introduced normalization based on an odds ratio (CP divided by well-known FIB), which helped to control for temporal changes in-sewer conditions and improved prediction accuracy (Fig. 3c).

Overall, the AI-derived framework developed in this study exhibits significant promise for use in surveillance, representing a non-invasive, unbiased, and cost-effective method. Collecting urban data (crowd-sourced, meteorological, and wastewater information) is non-invasive and anonymous, overcoming barriers associated with individuals who may resist disclosing their diagnosis due to the stigma involved. In addition, conventional clinical testing often relies on biased subsets of individuals, which may not accurately represent the entire population. The bias can result from variation in asymptomatic cases, differences in health-seeking behavior, and disparities in access to testing. A recent study suggests that pathogen counts in wastewater are indicative of both symptomatic and asymptomatic infections (Xiao et al., 2022). The wastewater microbiome data used in this study can be used effectively infer both symptomatic and asymptomatic cases within a sewershed area, providing an unbiased view of the prevalence of a disease. It is worth noting that the three WWTPs monitored in this study were located in Gyeonggi Province, Korea, which surrounds the capital city Seoul. The AI modeling thus benefitted from the data-rich environment, with Seoul and Gyeonggi accounting for about half of the country's population and two-thirds of infectious disease burden (e.g., COVID-19) (Yang et al., 2022), which helped to predict actual disease cases at a national scale. AI modeling also offers a cost-effective means for disease surveillance, particularly in urban areas, without the need for the invasive and expensive experimental testing of individuals. It leverages readily available and free environmental and crowdsourced data, making it an attractive option. However, it is essential to recognize that there are additional costs associated with acquiring wastewater samples and conducting sequencing, which would typically be borne by local administrative bodies.

Governments and local authorities already allocate resources for the regular inspection of foodborne disease surveillance in various sectors, such as agriculture, livestock, fisheries, and food processing environments (e.g., factories and restaurants). By law, restaurants must be visited by inspectors for sanitation monitoring at least once a year in countries such as Korea and the US (Sadilek et al., 2018). Accordingly, our study suggests the AI system should not replace existing conventional surveillance practices but rather be developed independently as a complement to them. We propose the inclusion of wastewater and other food and environmental samples in routine and proactive inspections for foodborne illness surveillance in the future. The digital surveillance system based on urban big data proposed in this study has the potential to provide an additional option that can strengthen and complement the

functions of the existing healthcare infrastructure and system, synergistically assisting policymakers and regulatory agencies in the public health sector.

5. Conclusions

- The present study introduced a ML modeling framework designed for the surveillance of foodborne infectious diseases, leveraging an integrative dataset derived from WW microbiomes, crowdsourced data, and environmental sources.
- The surveillance accuracy of the ML-based modeling framework was notably improved through systematic optimization of data integration, data normalization, selection of top-performing models, and hyperparameter tuning within the framework.
- XAI techniques employed in the framework quantitatively ranked robust input predictors (e.g., the abundance of the disease-causing agent in WW) based on their feature importance, influencing corresponding predictive surveillance decisions. This was validated through two independent feature sensitivity analyses employing different ML models.
- Multi-omics analyses, utilizing metagenomic and metatranscriptomic datasets, consistently recovered sequences informing the genotypes/toxinotypes of the disease-causing agent, thereby validating the ML-based surveillance results.
- The ML modeling framework, coupled with XAI methods and utilizing urban big data, developed in this study, facilitates data-driven digital surveillance for infectious diseases. This holds promise as a complementary and reinforcing tool for existing healthcare infrastructure and systems.

Data availability

The raw sequences have been deposited in the NCBI Sequence Reads Archive with the BioProject numbers PRJNA393385 (16S rRNA gene sequences), and PRJNA941081 (metagenome and metatranscriptome sequences).

CRediT authorship contribution statement

Seungdae Oh: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Haeil Byeon:** Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Jonathan Wijaya:** Visualization, Validation, Software, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2024.122282.

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